

# RELTWIN: LEARNING QUERY-SPECIFIC WINDOWS FROM CO-OCCURRING INVERSE RELATIONS

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## ABSTRACT

Audio temporal grounding must distinguish occurrences that contain the same sounds in different orders. We present RelTwin, which places inverse relations in one recording and contrasts their timestamp answers during training. Each competing window is acoustically valid but answers the opposite query, providing a locally confusable answer negative. Candidate cross-entropy augments sequence supervision while preserving the original inference procedure. In a same-environment, equal-update seed-0 comparison, RelTwin improves synthetic relation-development mIoU by 13.97 points, joint localization accuracy by 31.25 points, and full SpotSound-Bench mIoU by 1.02 points over SFT. On timing-modified recordings, its joint localization advantage remains 20.00 points. Across three seeds, RelTwin averages 59.16 SpotSound mIoU, exceeding the cited prior main-table best of 57.9. These results demonstrate the value of query-specific answer discrimination for generative audio grounding.

**Index Terms**— Audio temporal grounding, compositional reasoning, hard answer negatives, paired supervision

## 1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query-window binding error*. We target this distinction between event detection and query-specific localization.

Grounding draws on weak-caption learning [3], audio-language interaction in Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6], temporal representations in TimeAudio [7], and interleaved timestamps with absent-event training in SpotSound [8]. We focus on competition between *present* events: a plausible answer can contain the correct sounds

in the wrong relation. This requires explicit comparison of competing local answers.

CLAP learns audio-text matching [9]; T-CLAP combines concatenated sounds, reversed descriptions, and temporal contrastive training [10]. CompA examines order and binding [11], CoSTALA learns hierarchical spatio-temporal alignment [12], and SHINE uses compositional negative queries for video grounding [13]. RelTwin instead compares local timestamp answers: both inverse descriptions are true in one recording, and the negative window answers the opposite query.

RelTwin (Fig. 1) constructs opposite-order windows from shared excerpts. Candidate cross-entropy supplements sequence supervision, rewarding the requested query-window correspondence. Sharing the recording and categories focuses competition on the occurrence that answers the relation. Training compares candidates; inference retains timestamp generation.

Our contributions connect locally confusable answer construction, candidate-supervised adaptation, and a paired diagnostic requiring both queries to select their own targets. The SFT comparison shows gains of 31.25 joint-localization points and 1.02 SpotSound mIoU points. A timing intervention retains a 20.00-point joint-localization advantage. Together, these experiments demonstrate the value of query-specific answer discrimination.

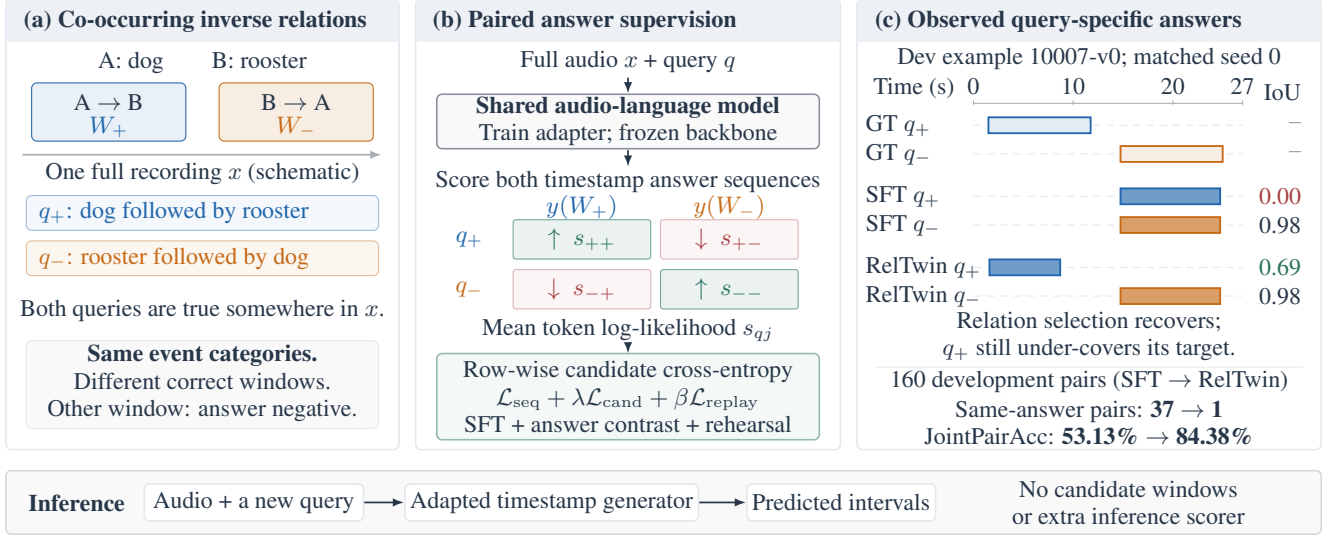
## 2. QUERY-SPECIFIC WINDOW LEARNING

### 2.1. Two true relations, two different answers

RelTwin reuses two ESC-50 excerpts [14],  $A$  and  $B$ , to form windows  $W_+ = (A \rightarrow B)$  and  $W_- = (B \rightarrow A)$  in one recording  $x$ . Each spans the *entire ordered sequence*. For inverse queries ( $q_+$ ,  $q_-$ ) and candidates ( $W_+$ ,  $W_-$ ), the answer-label matrix is  $Y = I_2$  (Fig. 1). Both queries are present at recording level; the paired labels specify their local correspondence.

Each recording has two templates, “followed by” and “and then,” in both directions. Category queries target all occurrences of  $A$  or  $B$  for rehearsal. Alternating the earlier relation controls a constant-position preference; excerpt reuse fixes source material within a pair. Section 4.2 tests

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**Fig. 1.** RelTwin overview. (a) Co-occurring inverse relations supply locally confusable answers. (b) The shared model contrasts timestamp sequences during training; arrows denote target preferences. (c) A cached seed-0 example (queries 56–57) shows relation recovery and boundary under-coverage. Aggregate statistics cover all 160 development pairs. The bottom path shows candidate-free timestamp generation at inference.

the contribution of layout-specific silence through a timing intervention.

## 2.2. Contrast timestamp answers during training

For the timestamp token sequence  $y(S)$  describing interval set  $S$ , define its mean token log-likelihood

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

Candidates are  $C = (W_+, W_-)$ . Their relative answer preferences are  $p_q(j) = \text{softmax}_j(s_\theta(x, q, C_j)/\tau)$ . The paired candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_{q_+}(1) + \log p_{q_-}(2)]. \quad (2)$$

Reversing the query exchanges the positive timestamp sequence. Candidate cross-entropy makes this local discrimination explicit alongside correct-answer sequence supervision and category-query rehearsal:

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Equation 3 defines *RelTwin*. SFT uses identical data and rehearsal with  $\lambda = 0$ . At inference, the model generates timestamps from audio and a query, without candidates or an additional scorer.

## 2.3. Joint localization and relation selection

For interval unions,  $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ . PairAcc requires both inverse-query predictions to achieve set-IoU  $\geq$

**Table 1.** Core training comparison (%). The seed-0 SFT/RelTwin pair shares training environment and updates. The backbone re-evaluation and three-seed summary provide separate references ( $\pm$ : sample SD). Joint: JointPairAcc.

| Method                 | SpotSound<br>mIoU | Relation development<br>mIoU | Joint  |
|------------------------|-------------------|------------------------------|--------|
| Current SFT, s0        | 58.42             | 74.58                        | 53.13  |
| RelTwin, s0            | 59.43             | 88.55                        | 84.38  |
| RelTwin – SFT, s0      | +1.02             | +13.97                       | +31.25 |
| SpotSound-A (re-eval.) | 58.42             | 37.64                        | 0.63   |
| RelTwin, 3 seeds       | 59.16 $\pm$ 0.26  | 87.05 $\pm$ 1.43             | 81.46  |

0.5. We additionally require each to overlap its own target strictly more than the opposite target, yielding JointPairAcc. Swap error means that at least one query violates this strict preference, including ties.

Returning the union of two disjoint equal-length targets for *both* queries gives IoU 0.5 each: PairAcc passes but JointPairAcc fails. Requiring a preference for the query’s own target distinguishes relational selection from broad temporal coverage. Inspired by paired compositional evaluation [15], we report JointPairAcc alongside standard IoU to capture this behavior.

## 3. EXPERIMENTAL SETUP

**Data and split roles.** Adaptation uses 40 ESC-50 classes and 256 constructed recordings: 1,024 relation queries (512 pairs)

**Table 2.** SpotSound-Bench results (400 queries, %). Published baselines: SpotSound Table 3 [8]. RelTwin reports three-seed means;  $\Delta$  compares with the cited prior main-table best for each metric. Bold marks the highest listed point estimate.

| Method                  | mIoU         | R1@.3        | R1@.5        |
|-------------------------|--------------|--------------|--------------|
| WTATG                   | 38.4         | 53.0         | 34.5         |
| AM-DETR                 | 19.5         | 25.5         | 15.0         |
| Gemini-2.5-Flash        | 25.7         | 34.8         | 27.8         |
| Gemini-2.5-Pro          | 19.7         | 20.0         | 17.3         |
| Kimi-Audio              | 3.2          | 3.0          | 1.3          |
| Qwen2-Audio             | 4.4          | 5.5          | 1.8          |
| Audio Flamingo 3        | 7.6          | 8.8          | 2.8          |
| TimeAudio               | 10.2         | 7.8          | 1.8          |
| SpotSound-Q             | 49.9         | 65.0         | 51.3         |
| SpotSound-A             | 57.9         | 76.5         | 59.5         |
| RelTwin (3 seeds)       | <b>59.16</b> | <b>77.17</b> | <b>62.08</b> |
| $\Delta$ vs. prior best | +1.26        | +0.67        | +2.58        |

and 512 rehearsal queries. The remaining 10 classes provide 80 recordings, 320 queries, and 160 inverse pairs. Adaptation/development classes and original Freesound IDs are disjoint; this separation is defined at the adaptation stage. Public evaluation covers all 400 SpotSound queries over 387 audios. The relation split and public scores have informed project development. We use the constructed split to diagnose query-window binding and SpotSound to measure performance on public grounding queries.

**Adaptation and the primary control.** Both methods retain SpotSound-A’s LoRA adaptation [16] on Audio Flamingo 3, with rank 8, scaling parameter  $\alpha = 16$ , dropout 0.1, and 20,185,088 trainable adapter parameters. SFT and RelTwin use 256 AdamW [17] updates, zero weight decay, learning rate  $5 \times 10^{-6}$ , gradient clipping 1.0,  $\tau = 1$ , and  $\beta = 0.5$ ; RelTwin uses  $\lambda = 1$ . Each update processes one inverse-query pair (two relation queries) and one category-rehearsal query. Gradients accumulate over these components, with no accumulation across pairs. The primary SFT–RelTwin comparison uses seed 0 with the same software environment, code, initial weights, and data order. Updates are matched; FLOPs and wall time are not. RelTwin scores four query–answer combinations without gradients then recomputes weighted gradients, whereas SFT backpropagates two positive answers. Both add rehearsal.

We also report RelTwin seeds 0/1/2, using final checkpoints and including every seed. SpotSound-A is re-evaluated with the same inference code as the adapted models to provide a backbone reference. All evaluations use 16 kHz mono audio, bfloat16 inference, and greedy decoding (one beam, no sampling) with at most 128 new tokens per query. The three-seed RelTwin summary is separate from the matched seed-0 SFT comparison.

**Uncertainty and reproducibility.** We verify row alignment, audio identity, duration, and recomputed IoU. Bootstrap uses 20,000 audio-group draws, keeping queries together. Relation CIs additionally resample 39 original-source connected groups (largest: 18 audios) to address excerpt reuse. The intervals quantify sample uncertainty conditional on observed seeds and do not adjust for development selection. Code, inputs, weights, and environment are hashed before inference.

## 4. RESULTS AND ANALYSIS

### 4.1. Candidate supervision improves binding

RelTwin improves both relation selection and temporal overlap in the primary SFT comparison (Table 1). Relation mIoU increases from 74.58 to 88.55 (+13.97 points), with source-group 95% CI [9.84, 17.51]. JointPairAcc rises from 53.13 to 84.38 (+31.25 points; CI [22.35, 40.50]), and Swap error falls from 40.00 to 7.50. On SpotSound, SFT scores 58.42 and RelTwin 59.43: +1.02 mIoU points with audio-group CI [0.23, 1.81]. Shared positive examples and update counts connect these gains to adding explicit competition between timestamp answers. Across the 400 public queries, the paired win/tie/loss counts are 92/239/69, including two IoU decreases of at least 0.5.

Paired diagnosis reveals a change in answer selection. SFT returns identical answers for 37/160 inverse-query pairs (22 audios), compared with 1/160 for RelTwin; 27 of those 37 pairs become jointly correct. Overall, 51 pairs change from incorrect to correct, one in the opposite direction, and 25 remain incorrect. Figure 1 shows a cached recovery case: the selected window answers the requested order, while under-coverage exposes the separate role of boundary accuracy. JointPairAcc ties the changed answers to the correct relation targets.

### 4.2. Binding gains persist after timing intervention

The timing intervention tests whether layout-specific silence explains the binding advantage. We recompose all 80 development audios with 1.5 s lead-in, 0.25 s within-window gap, 3 s between windows, and 2 s tail, preserving event waveforms, order, and queries. Frozen SFT/RelTwin seed 0 models evaluate one template (160 queries). Forty AB-first audios remain waveform-identical and reproduce every prediction; the 40 BA-first audios receive the changed timing layout.

On the changed audios, RelTwin achieves 87.50 JointPairAcc versus 67.50 for SFT: +20.00 points, source-group CI [2.94, 33.33]. Under the original layout, the corresponding scores are 85.00 and 42.50. Uniform timing raises SFT’s score and reduces the gap from 42.50 to 20.00 points, while the positive interval shows a retained binding advantage. Across all 160 queries, uniform-timing mIoU is 81.34 for SFT and 90.65 for RelTwin. This targeted, post-development

test shows that the original silence pattern does not fully explain the answer-discrimination advantage.

### 4.3. Public grounding performance

RelTwin averages 59.16 SpotSound mIoU across three seeds, exceeding the cited prior main-table best of 57.9 by 1.26 points (Table 2). R1@.3 and R1@.5 rise by 0.67 and 2.58 points, linking answer discrimination to overlap and recall gains on public queries. Literature scores provide performance references; Table 1 supplies the SFT contrast. Against the current official re-evaluation at 58.42 mIoU, RelTwin gains 0.75 points, with audio-group CI  $[-0.24, 1.74]$  spanning zero.

### 4.4. Scope

Repeated excerpts and fixed templates enable controlled binding tests. Alternative hard-negative constructions, matched multi-seed replication, and natural overlapping or long-range relations remain open evaluations. Adaptation-stage source separation does not audit backbone pretraining overlap.

## 5. CONCLUSION

RelTwin contrasts valid windows that answer different relation queries. Candidate supervision improves joint localization and SpotSound grounding over the paired SFT control, retaining an advantage under timing intervention and preserving the original inference procedure. Competition between locally confusable timestamp answers provides a practical training strategy for query-specific audio grounding.

## 6. ACKNOWLEDGMENTS

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## 7. COMPLIANCE WITH ETHICAL STANDARDS

This computational study reuses public audio from ESC-50 [14] and SpotSound-Bench [8], including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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